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Drivers of plant community composition and diversity in low Arctic western Greenland

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The Arctic experiences rapid climate change, but our ability to predict how this will influence plant communities is hampered by a lack of data on the extent to which different species are associated with particular environmental conditions, how these conditions are interlinked, and how they will change in coming years. Increasing temperatures may negatively affect plants associated with cold areas due to increased competition with warm-adapted species, but less so if local temperature variability is larger than the expected increase. Here we studied the potential drivers of vegetation composition and species richness along coast to inland and altitudinal gradients by the Nuuk fjord in western Greenland using hierarchical modelling of species communities (HMSC) and linear mixed models. Community composition was more strongly associated with random variability at intermediate spatial scales (among plot groups 500 m apart) than with large-scale variability in summer temperature, altitude or soil moisture, and the variation in community composition along the fjord was small. Species richness was related to plant cover, altitude and slope steepness, which explained 42% of the variation, but not to summer temperature. Jointly, this suggests that the direct effect of climate change will be weak, and that many species are associated with microhabitat variability. However, species richness peaked at intermediate cover, suggesting that an increase in plant cover under warming climatic conditions may lead to decreasing plant diversity.

Keywords: Arctic vegetation, climate impact, low Arctic tundra, microhabitat variability, plant–plant interactions, species richness, vegetation composition

Introduction

The Arctic is experiencing rapid warming and altered precipitation patterns, which has already had widespread effects on plant communities (Elmendorf et al. 2012a, Heijmans et al. 2022). In many areas tall shrubs have responded to warming by growing

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faster and becoming increasingly abundant (Walker et al. 2006, Myers-Smith et al. 2015, Mekonnen et al. 2021), but effects vary among species groups (Elmendorf et al. 2012b). Species associated with late melting snow patches are expected to respond negatively to climate change, as this habitat will gradually disappear when temperatures increase (Le Roux et al. 2013b, Nabe-Nielsen et al. 2017), and slow growing species adapted to harsh and cold conditions may suffer from more intense competition when climate change permits more competitive, warm-affiliated species to become increasingly abundant (Elmendorf et al. 2012b, Oke et al. 2023). Intensified competition and disappearance of species associated with snow patches will result in decreasing species richness unless counterbalanced through immigration of new species (Niittynen et al. 2018, Robinson et al. 2018, Rixen et al. 2022). To predict the long-term consequences of climate change on community composition and species richness it is therefore essential to understand whether plant communities are predominantly structured by factors that will be influenced by climate change or by factors that remain nearly constant through time.

Although the effects of climate change have been documented in several long-term studies from different parts of the Arctic, the patterns that emerge are highly variable (Myers-Smith et al. 2019, Bjorkman et al. 2020). While some studies have documented an increase in shrub growth, responses are heterogeneous (Elmendorf et al. 2012a, Myers-Smith et al. 2019, Bjorkman et al. 2020, Mekonnen et al. 2021). Graminoids have also become increasingly abundant in some areas, although the relative abundance of different plant functional groups has been relatively unaffected by recent climate change (Bjorkman et al. 2020). One way to obtain additional insights into the influence of future climate change is to use space-for-time substitutions based on vegetation studies covering climatic gradients that resemble those that we expect to find in the future. Such studies assume that climate–biotic relationships observed in space can faithfully predict changes that occur over time (Lovell et al. 2023). The high geographical variability in plant community responses to environmental conditions underlines the importance of establishing such studies across the Arctic to improve our understanding of how climate impacts are moderated by other environmental factors.

Another thing that makes predictions of climate change effects challenging is that plants are influenced by numerous environmental factors that operate at different spatial and temporal scales (Fig. 1). Broad-scale effects of warming interact with fine-scale variations in snow cover and soil conditions, which are related to local topography (Martin et al. 2017, Kemppinen et al. 2018, Niittynen et al. 2020). The effects of climate warming also depend on local soil water conditions (Le Roux et al. 2013b, Winkler et al. 2016), and these are in turn affected by precipitation and hydrology (Taylor and Seastedt 1994). Temperatures and wind both affect species distributions and vary with altitude (Billings 1973, Körner 2003, p. 200). Thus, although plants are influenced by large-scale climatic variations, such as those that occur along altitudinal or coast-to-inland gradients (Prendin et al. 2022), these effects are often masked by local variations in other environmental factors (Suggitt et al. 2011, Graae et al. 2017, Rissanen et al. 2023). To draw general conclusions about the relative impact of different drivers it is therefore important

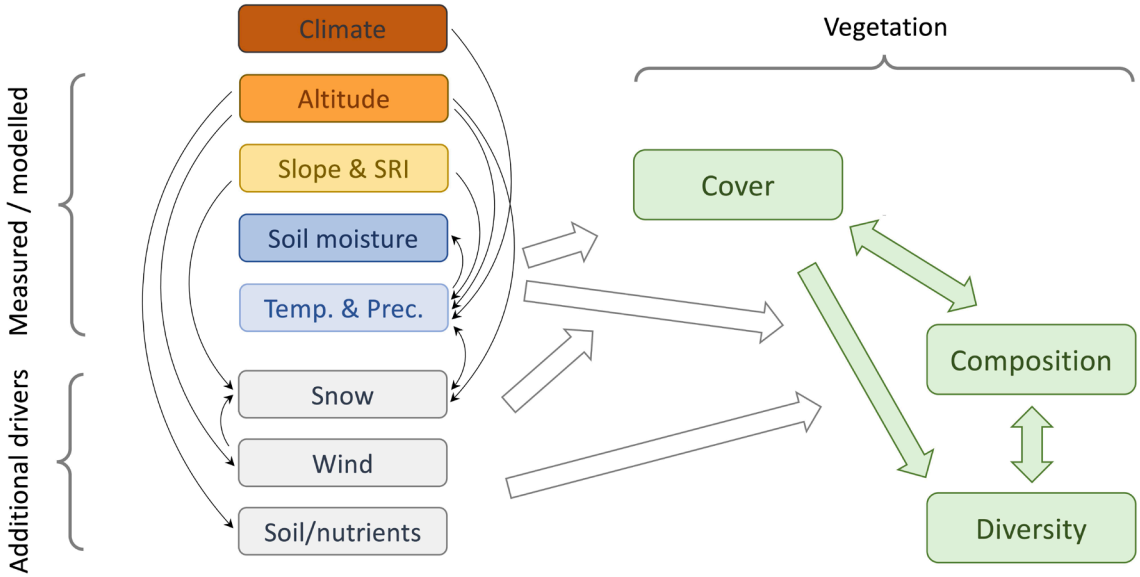


Figure 1. Interactions between potential drivers of Arctic plant community composition investigated in this study. Altitude, slope and solar radiation (SRI) are invariant in time. Climate change is expected to predominately influence temperature (temp.) and precipitation (prec.). White arrows show direct and indirect impacts of abiotic drivers on plant communities, and green arrows indicate effects of plant–plant interactions.

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studying fine-scale variations in plant communities across naturally occurring environmental gradients.

One interesting consequence of having multiple interacting drivers of plant growth is that fine-scale environmental variability may buffer against effects of climate change, or at least postpone them (Suggitt et al. 2018, Huxley and Spasojevic 2021). Temperature changes after decades of climate warming are in many cases lower than variations among microsites with different topography and hydrology, making it possible for species to escape the effects of climate change by relocating locally (Scherrer and Körner 2011, Opedal et al. 2015, Jiménez-Alfaro et al. 2024). When fine-scale temperature variability is related to occurrence of late-melting snow patches or streams originating from glaciers, local extinctions may occur with a delay when these habitats gradually disappear and both temperature and soil moisture variability diminish (Rissanen et al. 2023). In either case climate change may ultimately cause local plant population sizes to decline, potentially resulting in decreased local diversity.

Biotic interactions, and particularly facilitation and competition, can also play a key role in structuring Arctic plant communities (Dormann and Brooker 2002). Recruitment facilitation has been demonstrated to promote species coexistence in harsh environments, and in more benign environments a dense shrub layer can prevent new species from establishing and result in competitive exclusion of shade-intolerant species (Elmendorf et al. 2012b, Mekonnen et al. 2021). The rate at which Arctic plant communities respond to climate change therefore partly depends on new plant species' ability to establish and persist in already established plant communities, which can result in considerable time lags in community responses (Svenning and Sandel 2013, Essl et al. 2015). The intensity of these biotic interactions will thus have a strong impact on how rapidly plant communities respond to climate change.

The objectives of this study were to determine whether plant species richness and community composition were related to large-scale climatic variability, fine-scale abiotic drivers or plant–plant interactions in the Nuuk region in western Greenland. This was done based on vegetation data from 414 permanent plots placed in five sites along the Nuuk fjord (Godthåbsfjorden) using a stratified random design. Due to the complex topography in this region, we expected fine-scale variations in slope, soil moisture and solar radiation to be more important than broad-scale climatic variability, although we did expect that differences among sites would be sufficient to affect community composition. We further hypothesized that species richness would be highest at intermediate plant cover where competition is not too severe, but where conditions are sufficiently benign to allow many species to survive. At the same time, we expected species with inferior competitive abilities to occur less often in areas with high cover. This is one of the largest studies of the factors that determine plant community composition and species richness in Greenland, and it will thus improve our understanding of how Arctic plant communities respond to climate change in different geographical regions.

Material and methods

Study site

The study was conducted at five different sites along the Nuuk fjord (Godthåbsfjorden) in western Greenland (Fig. 2). The fjord stretches from the Davis Strait in the west and 160 km eastward towards the inland ice. The mean annual air temperature in Nuuk (64°10'N, 51°44'W) was 1.0°C in the period 1990–2020 and the mean annual precipitation was 847 mm (Cappelen et al. 2021). The five study sites were selected prior to the field work along the entire fjord to assess how the degree of continentality affected the plant species composition. They were selected to be approximately evenly distributed while avoiding places with steep cliffs along the shore.

The vegetation is dominated by different species of shrubs. In the western end of the fjord *Empetrum nigrum* is particularly common, whereas *Betula nana and Salix glauca* gradually become more abundant further east. Here patches of *Ledum groenlandicum* and *Alnus crispa* also occur in some places, and both *A. crispa* and *S. glauca* grow to become >1 m tall in the eastern end of the fjord (Supporting information). Bogs with *Carex saxatilis* and *Eriophorum angustifolium* are widespread. At higher altitudes (> 300 m a.s.l.) the vegetation gradually becomes more open, particularly on steep rocky slopes.

Study design

In 2011–2013, 414 permanent plots were established across the five study sites. The plots were circular, with a diameter of 2 m. Three groups of six plots were selected for every 100 m increase in altitude. The position of the first plot at each altitude was selected by walking uphill straight towards a preselected point until the isocline was reached (measured using a hand-held GPS). Plot centres were placed exactly 10 m apart along the isoclines, or slightly more if needed to prevent plots from being entirely in water or having an average slope > 45°. Plot groups were placed exactly 500 m apart. The lowest plots were moved to 20 m a.s.l. to avoid exposure to salt. The highest altitudes used in sites 1–5 were 200, 200, 400, 500 and 500 m a.s.l., respectively, reflecting that the mountains are higher further east. The study design was the exact same as previously used in Young Sund (Nabe-Nielsen et al. 2017).

The vegetation survey included a complete inventory of all species of vascular plants in each plot. This inventory was conducted in 2011–2013. Maximum height and diameter were measured for the woody species in 2021–2022. The cover of woody species, graminoids, and herbaceous plants was assessed for each plot by two independent observers as percent cover < 2 m from plot centres. Subsequently the two estimates were averaged. Plants were named following Böcher et al. (1978).

Environmental variables

To assess if plant species richness or community composition were related to fine-scale variations in soil conditions and

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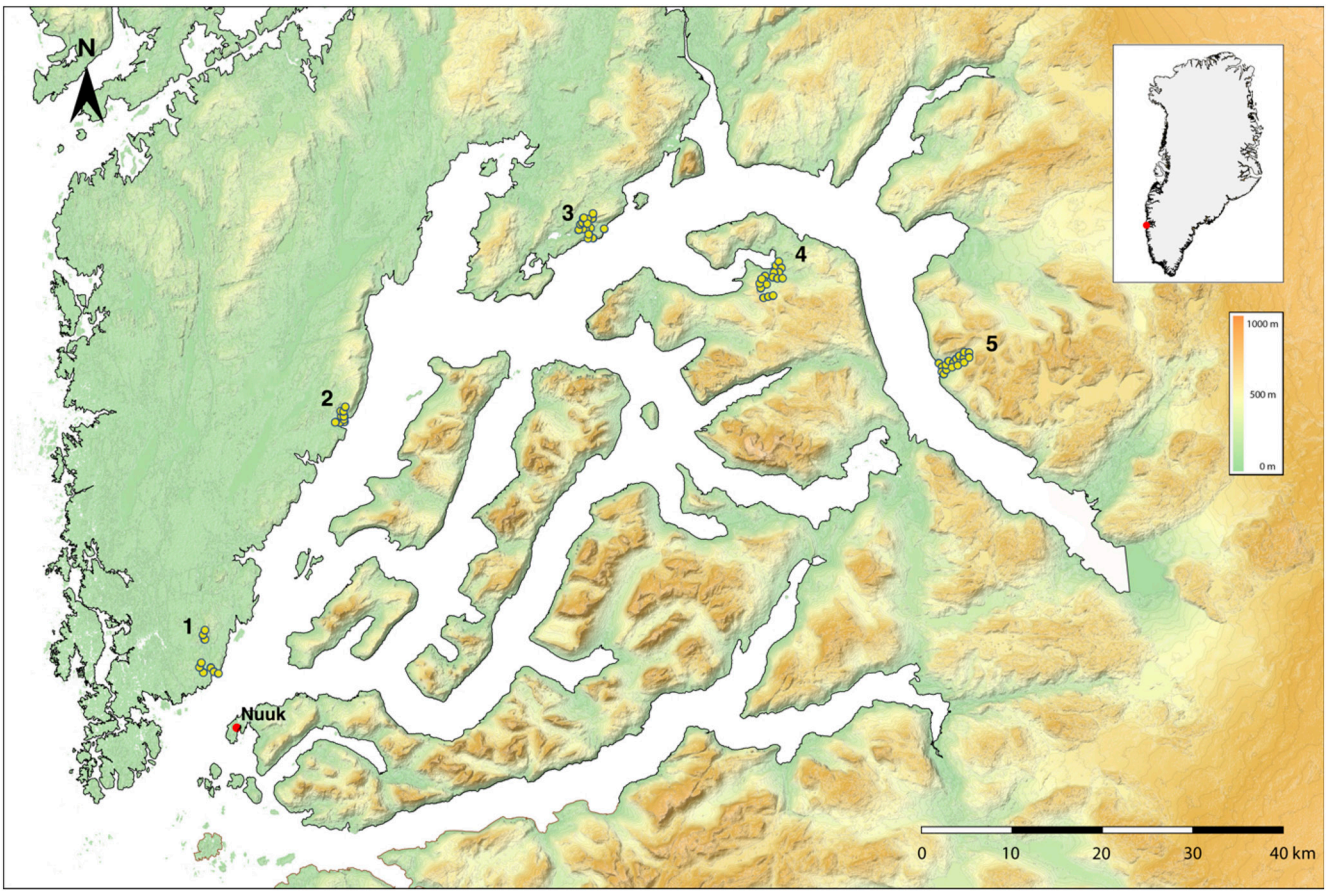


Figure 2. Map of the five study sites along the Nuuk fjord in western Greenland. Each of the 69 yellow dots represents a plot group consisting of six vegetation plots. The inset shows Greenland with the red dot indicating the location of Nuuk.

topography, we collected data on multiple environmental variables for each plot. The geographic position was recorded using a handheld GPS (Garmin GPSmap 62s); the slope of the terrain and the direction of the slope were recorded using a handheld clinometer and a compass. Based on these variables, we computed the solar radiation index (SRI; Keating et al. 2007). The area covered by rocks or unvegetated stones with diameter > 2 cm was scored for each plot. Soil moisture was recorded in three places 1-m from the centre of each plot using a ML3 Theta Probe (Delta-T Devices), and values were averaged. In 19 of the 414 plots soil moisture could not be measured; here the average soil moisture from the entire plot group was used.

To assess the influence of climate on the plant communities, we used monthly temperature and precipitation data from the CHELSA database ver. 2.1 (Karger et al. 2017). Temperatures were based on a statistical downscaling of atmospheric temperatures, while precipitation was modelled based on wind fields, exposition, boundary layer height and other orographic predictors. Data were available at a spatial resolution of 30 arc-s (approx. 1 km). We extracted values of temperature and precipitation corresponding to the plot locations and neighbouring grid cells up to 250 m away and used the average of these in subsequent analyses. In the statistical analyses we used average mean monthly temperature and precipitation for the months June–August of 2012 to characterize climate in the plot groups during the peak growing season. In addition, we used precipitation for the months November–January as a measure of snow cover, which might affect the onset of the growing season. In June–August mean temperatures were in the range 7–16°C in all plot groups (Supporting information) and the monthly precipitation ranged from 20–280 mm.

Testing for possible drivers of species richness

To determine how number of plant species per plot group was related to environmental variability, we fitted a linear mixed model with average altitude, slope, SRI, soil moisture, summer temperature and vegetation cover per plot group as fixed effects and site as random grouping variable. To test if species richness peaked at intermediate plant densities, plant cover was included as a second order polynomial. Summer temperature was strongly negatively correlated with winter and summer precipitation (Supporting information), so to avoid multicollinearity these variables were not included in the model. For the remaining predictors the variance inflation factors (VIF) were < 2.5 (Table 1), indicating low levels

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Table 1. Linear mixed models explaining variation in number of species per plot group or average vegetation cover in plot groups. Here sw shows the relative importance of predictor variables and VIF are variance inflation factors. R^2 is provided for fixed effects (m) and the entire model (c).

Variable	F	p	sw	VIF
Response: species per plot group			$R^2_m = 0.48$	$R^2_c = 0.48$
Altitude	21.49	< 0.001	1.00	1.56
Slope	22.17	< 0.001	1.00	1.11
SRI	1.75	0.192	0.45	1.32
Soil moisture	2.19	0.145	0.51	1.46
Summer temp.	1.65	0.204	0.41	1.34
Cover	2.45	0.123	0.28	1.45
Cover sq.	3.79	0.056	0.25	1.17
Response: mean cover in plot group			$R^2_m = 0.27$	$R^2_c = 0.40$
Altitude	21.12	< 0.001	0.98	1.29
Slope	0.11	0.741	0.27	1.10
SRI	0.34	0.560	0.37	1.31
Soil moisture	4.15	0.046	0.51	1.28
Summer temp.	1.16	0.286	0.33	1.29

of collinearity (Johnston et al. 2018). In addition to assessing the significance of individual predictors, we calculated their relative importance using the 'MuMIn' package for R (Bartoń 2023). Analysis of residuals suggested that number of species was linearly related to all predictors (except plant cover), with no serious outliers. Residuals were approximately normally distributed. The mixed model was fitted using the 'nlme' package for R (Pinheiro and Bates 2023).

We tested how plant cover was influenced by altitude, slope, SRI, soil moisture or summer temperature using the same approach. Again, VIF were < 2.5 and cover was linearly related to the predictors.

To assess to what extent number of species per plot was related to variations among plot groups on the same altitude, variation among altitude bands or among sites, we performed a variance component analysis using the R package 'VCA', with plot group nested within altitude within site. All statistical analyses were performed using R ver. 4.4.1 (www.r-project.org).

Analysing community responses to environmental variation

We used hierarchical modelling of species communities (HMSC) to determine if plant community composition was related to altitude, slope, SRI, soil moisture, summer temperature and vegetation cover (Ovaskainen et al. 2017, Tikhonov et al. 2020). HMSC is a joint species distribution model that makes multivariate inference about environmental effects across many taxa (Warton et al. 2015). Unlike in the univariate linear mixed models described in the previous section, in HMSC the response variable is multivariate, modelling simultaneously the responses of each species to the environmental and latent predictors. Thus, while the univariate analyses concerned variation in species richness, the HMSC analyses are targeted to ask what drives variation in the occurrences of each of the individual species. In this analysis we modelled the occurrences of the 55 species that were found in at least 10 of the 414 plots (species list in the Supporting information). We built a probit regression of species occurrences using plant cover and the six environmental variables described above as fixed effects, and used plot, plot group and site as random effects. Here cover (total cover of all species in individual plots) was included as a second order polynomial.

We fitted the model using default prior distributions with the 'Hmsc' package for R (Tikhonov et al. 2020) and sampled the posterior distributions with four Markov chain Monte Carlo (MCMC) chains, each run for 375 000 iterations. The first 125 000 of these were removed as burn-in. The chains were thinned by 1000 to yield 250 posterior samples per chain and 1000 posterior samples in total. We considered MCMC convergence satisfactory, as the mean potential scale reduction factor (PSRF; Gelman and Rubin 1992) of the model parameters characterizing environmental responses was 1.02 and thus close to the ideal value of one.

We measured the explanatory powers of the model by Tjur's R^2 and AUC. To quantify the importance of the different fixed and random predictors, we performed a variance partitioning. To evaluate the species responses to each covariate, we examined if the posterior probability for each response to be positive (or negative) was at least 0.95.

To quantify the influence of cover for each species, we computed the posterior probabilities by which the species occurrence probability is predicted to be highest at low cover, intermediate cover, or high cover. To do so, we predicted for each posterior sample and for each species the cover c^* at which the occurrence probability reached its highest value. We summarized the distribution of c^* over the posterior samples by the posterior median of c^*, and by the posterior probabilities by which c^* is low ($c^* = 0$), intermediate ($0 < c^* < 100$) or high ($c^* = 100$).

Results

Potential drivers of plant species richness

Plant species richness was related to altitude and slope but not to soil moisture, solar radiation or summer temperature (Table 1). At the same time there was a close to significant effect of the quadratic term of plant cover, indicating that species richness was lower in areas with very dense or very scarce vegetation. The highest number of species was found in plot groups at high altitudes and on steep slopes (Fig. 3). This trend was the same in all study sites (Supporting information). The high-altitude plot groups at the same time tended to have more bare rock (Pearson $r = 0.13$), as were those on steep slopes ($r = 0.11$). Plot groups with dense plant cover, and hence with lower species richness, were more common in dry areas and at low altitudes (Table 1, Supporting information). The full model explained 48% of the variation in number of species per plot group, but a model that only included the most important predictors (altitude and slope)

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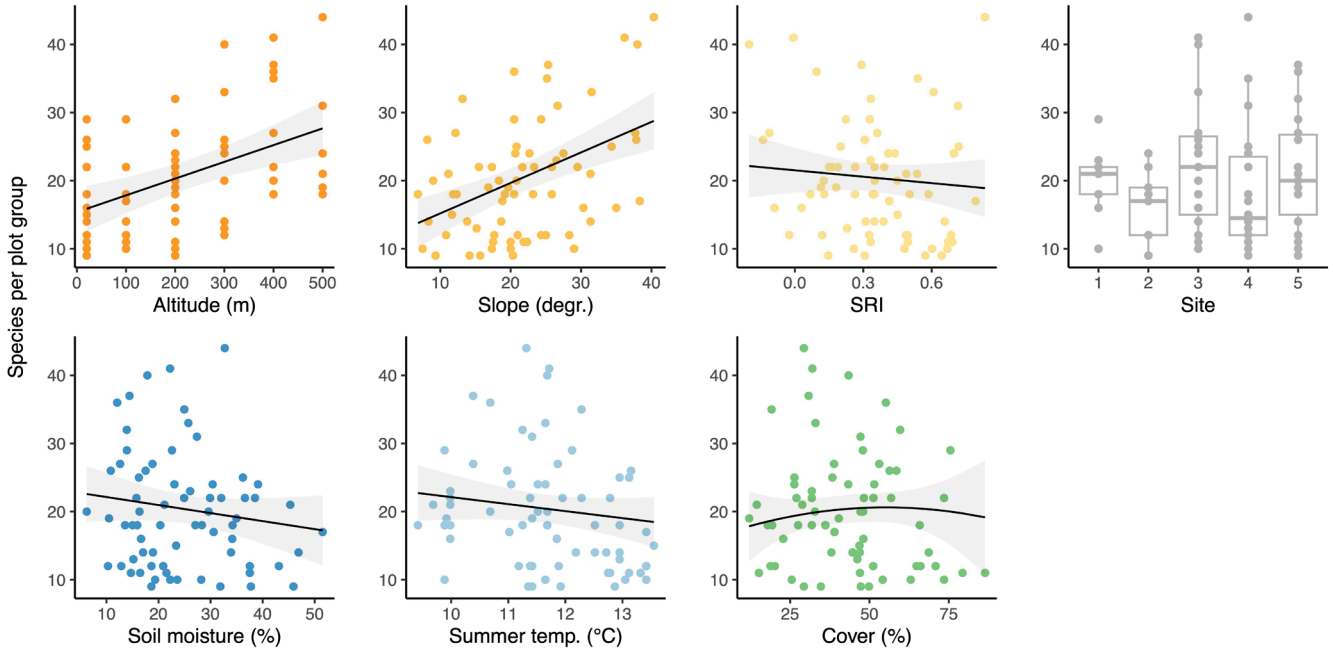


Figure 3. Relationship between number of species per plot group and environmental variables. Cover was fitted as second order polynomial. Shaded areas show confidence intervals based on the full model in Table 1.

was slightly better (ΔAIC = 3.0). In this model the fixed effects explained only 42% of the variation.

The variation in number of species per plot was predominately related to variations among plot groups (38.3%) and plots within plot groups (35.6%), and to a smaller extent related to variation among isoclines (18.3%) and sites (7.8%; Supporting information).

Plant community responses to environmental variation

The HMSC model achieved a high explanatory power, with average AUC = 0.93 and Tjur's R^2 = 0.34 among the species. Plant community composition was more strongly related to random differences among plot groups (11.2% of total variance in the model, or 28.8% of the explained variance; Fig. 4) than to variation among sites (2.3%) or to any of the environmental variables. The environmental variables and plant cover jointly explained 17.9% of total variance in species occurrences, with altitude (4.0%) and temperature (4.5%) being most important. For 12 of the 55 species in the HMSC analysis, random variation among plot groups explained >20% of the total variance. In contrast, none of the measured/modelled environmental variables explained >20% of the variation in occurrence, except for *Ledum groenlandicum* where temperature alone explained >23% of the variance.

Many of the species in the analysis were associated with high altitudes and steep slopes (Fig. 5), and among the most common species (in ≥ 100 plots, Supporting information) it was only *Betula nana*, *Carex bigelowii* and *Vaccinium uliginosum* that predominately occurred on flat ground. Species occurrence was only significantly related to vegetation cover for 12 of the 55 species (Supporting information). Four of the 12 species with at least 90% posterior support for cover were small-statured species associated with low cover (i.e. < 30% cover). Three species were associated with intermediate cover (30–55%), and five species were associated with high cover (100%). The amount of statistical support for an influence of cover on occurrence varied among species (Supporting information). Among the species where the influence of cover was best supported, all the species that predominately occurred at low cover were small statured (Supporting information).

Discussion

Drivers of species richness

Plant species richness was greater in high-altitude areas and on steep slopes along the Nuuk fjord, but it was not related to mean summer temperature, soil moisture or solar radiation (SRI). The influence of altitude and slope was the same across study sites (Supporting information), and jointly these variables explained a large part (42%) of the variation in number of species per plot group. However, the variation in number of species per plot was more strongly related to random variations among plot groups on the same altitude (38.3%) than to altitude itself (18.3%) or to variations among sites (7.8%), suggesting that local species richness is more strongly related to fine-scale environmental conditions than to large-scale scale climatic variation.

The two types of fine-scale environmental variation that are most likely to generate the observed differences among plot groups is snow cover and disturbances. Landslides

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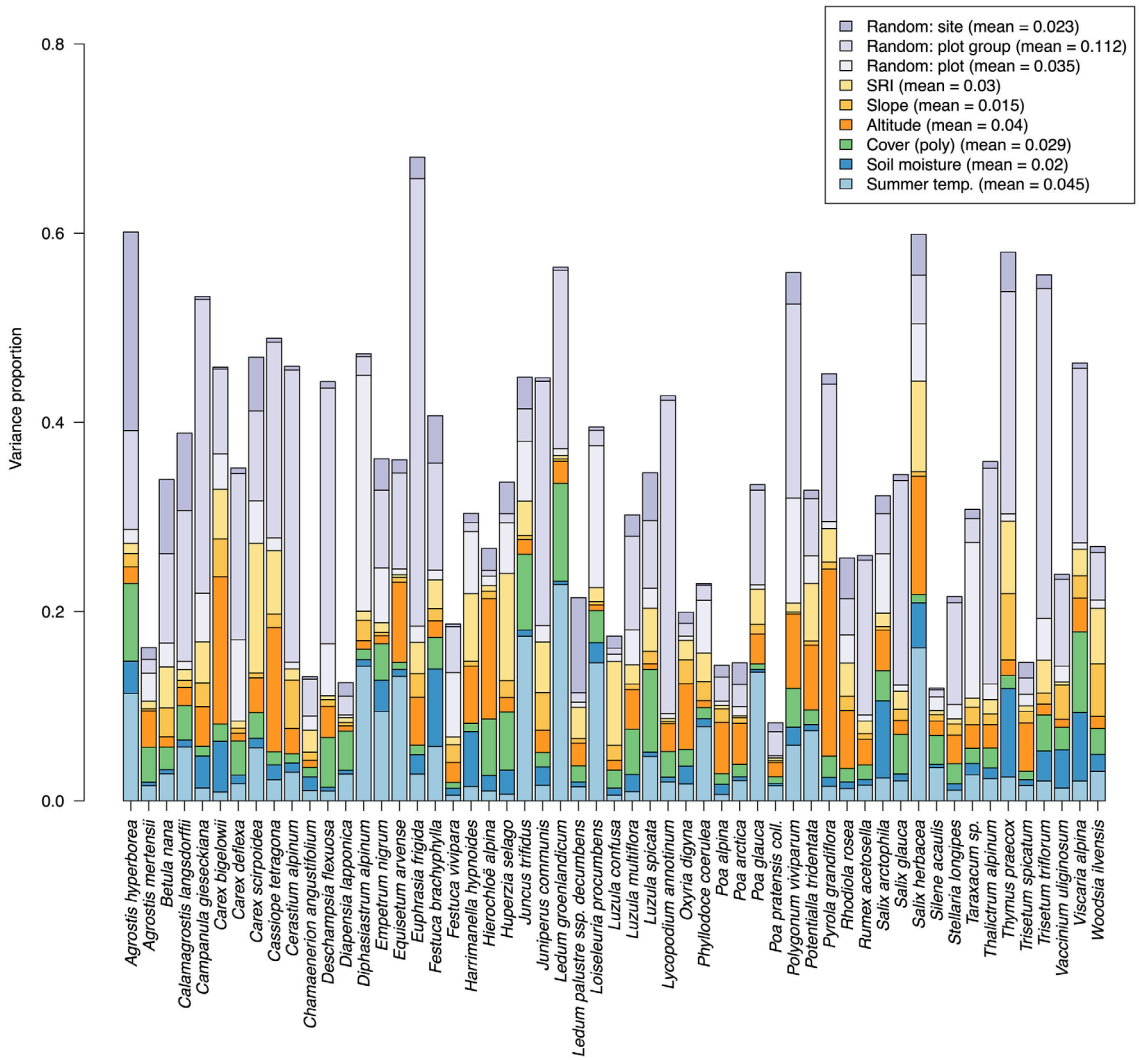


Figure 4. Proportion of variance in each species' occurrence explained by environmental conditions, plant cover and random differences among plots, plot groups and sites. Proportion of variance for cover refers to the combined variance for the linear and the quadratic term.

and fine-scale soil disturbances, such as when frost causes gravel to become unstable and slide down, tend to be more prevalent on steep slopes, creating a mosaic of microsites where both pioneer and late successional species can persist (Restrepo et al. 2009, Franklin 2010, Le Roux et al. 2013a). The higher prevalence of bare rock and unvegetated stones on steep slopes and in high-altitude areas suggests that this is also the case in the Nuuk fjord area. Similarly, snow tends to be more patchily distributed at higher altitudes where strong winds cause it to accumulate in depressions (Callaghan et al. 2011, Lehning et al. 2011) or along projecting rocks (Böcher 1954). This permits species that tolerate desiccating winds early in the growing season to coexist with species that benefit from being snow covered for extended periods, thus resulting in higher local diversity (Carlsson and Callaghan 1991, Rissanen et al. 2023). Such topographically mediated variations in snow cover can have strong impacts on how climate change influences the vegetation (Niittynen et al. 2020, Oldfather et al. 2023), and can to some extent be self-amplifying due to tall shrubs' tendency to accumulate snow (von Oppen et al. 2022).

Species richness also tended to be lower in plot groups with very high plant cover, although the tendency for richness to peak at intermediate cover was only marginally significant. This suggests that competition is particularly important in more productive areas (Michalet et al. 2006, Virtanen et al. 2013). These productive areas with dense vegetation were generally found at low altitudes, as also reported in other studies (Bruun et al. 2006).

Drivers of plant community composition

Plant community composition was more strongly related to random variations among plot groups than to any of the studied environmental variables, indicating that most species are found in areas that cannot be pinpointed based

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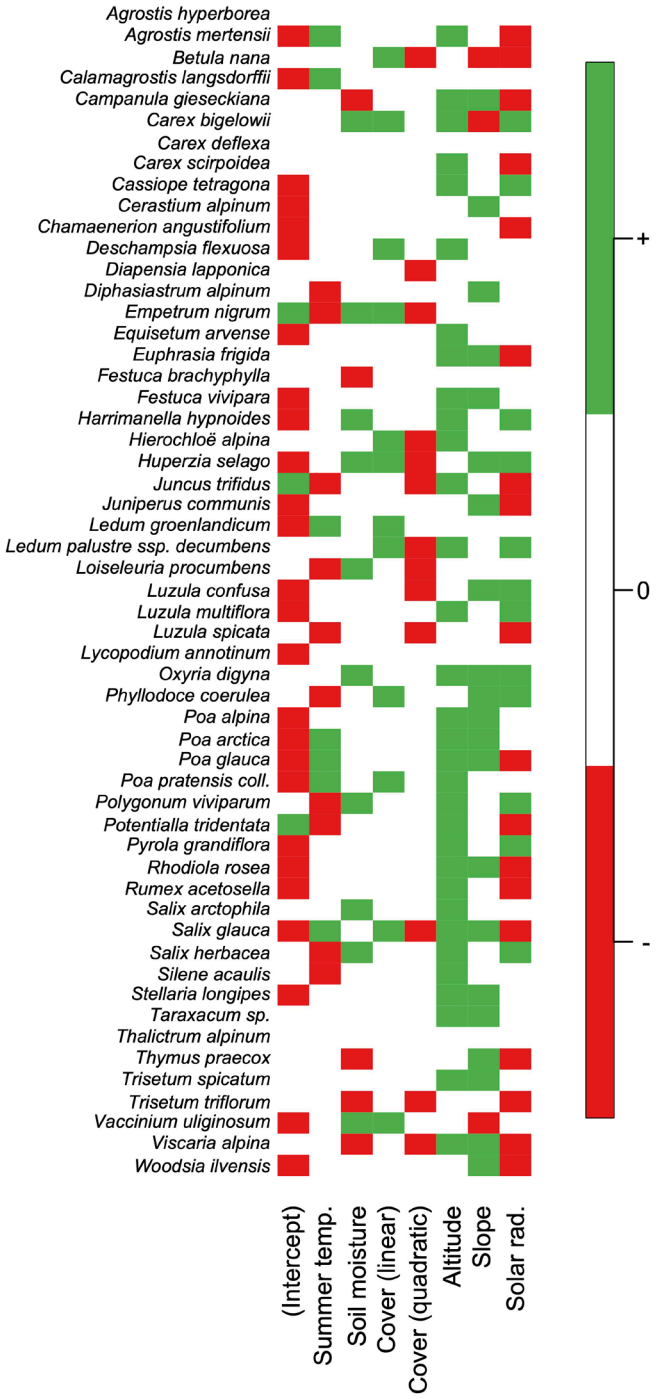


Figure 5. Statistically supported species-level responses to the predictors. Responses that were estimated to be positive with at least 95% posterior probability are shown by green, whereas those that were estimated to be negative with at least 95% posterior probability are shown by red. A negative quadratic cover term indicates that the species is most likely to occur at intermediate vegetation densities.

on broad-scale climatic variables or soil moisture (Fig. 4). Summer temperature was the most important climate-related variable, yet it explained on average only approximately 5% of the variation in species occurrences. Another study based on only the shrubs in the same data set pointed temperature variability out as important for several species of shrubs (von Oppen et al. 2021). This suggests that not only summer temperatures but also winter temperature minima can play a role for plant communities in the Nuuk area. The large variation among plot groups suggests that plant communities are predominately structured by variables related to fine-scale topography, including snow cover and soil disturbance, or to variations in, e.g. herbivory or nutrient availability (Martin et al. 2017, Kemppinen et al. 2018, Niittynen et al. 2020), and that large-scale climatic variability plays only a smaller role.

The weak relationship between environmental variability and plant community composition contrasts with findings from other parts of Greenland. In our study the variables altitude, slope, SRI, soil moisture and summer temperature jointly explained only 15.0% of the variation in plant community composition, whereas the same variables explained 29.4% of the variation in Young Sund in north-east Greenland (Nabe-Nielsen et al. 2017). This difference can partly be explained by a higher sensitivity to climatic variability for plants that live at the northern edge of their distribution (Myers-Smith et al. 2015). *Cassiope tetragona* is an example of a species with a distribution that is limited by temperature at its northern range (Havström et al. 1993), but whose distribution is much more strongly related to altitude than temperature in our study. Instead the prevalence of *C. tetragona* may be indirectly determined by nutrient availability in the southern parts of its range around the Nuuk fjord (Havström et al. 1993, Myers-Smith et al. 2015), as nutrients generally favour more competitive species.

Environmental variability may not be equally important for rare and abundant species, as species that can grow under a wide range of environmental conditions are more likely to become abundant. In our study the rare species (in < 10 plots, and therefore not included in the HMSC analyses) include *Carex nardina*, *C. supina*, *Saxifraga tricuspidata*, *Artemisia borealis*, *Gentiana nivalis* and *Kobresia myosuroides*, that have been labelled xerophytes on basis of their adaptations to living in dry conditions (Warming et al. 1895, Böcher 1954). These species were exclusively found on south-facing slopes at elevations ≥ 200 m, except at the site closest to the inland ice where they also occurred at lower altitudes. The rare species also include the hygrophytes *Gnaphalium supinum*, *Carex lachenalli* and *Sibbaldia procumbens* that were found at the site furthest from the inland ice and on north-facing slopes at altitudes above 400 m on sites in the middle of the fjord. The omission of such habitat specialists presumably caused the HMSC analysis to under-estimate the effect of climate-related variables on plant community composition. It may also be the reason why the analysis did not suggest a strong influence of plant cover on species occurrence, or that species with inferior competitive abilities tended to occur less often in areas with high cover. The rare species all appeared to be more climatically specialized than the more abundant species, yet, due to their rarity they play a disproportionate role for local and landscape-scale diversity.

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Long-term community responses to climate change

Plant community composition and species richness are not strongly related to average summer temperature in the Nuuk area, indicating that broad-scale temperature differences have played a minor role in structuring plant communities. Instead, their composition and richness are predominately influenced by fine-scale differences in soil conditions, exposure to winds, biotic interactions, and other factors that vary among plot groups. As many of these factors are likely to stay relatively constant through time, they can help buffering plant communities against direct impacts of climate change (Ackerly et al. 2010, Scherrer and Körner 2011, Nabe-Nielsen et al. 2017). Local topography also affects the degree of shading and local temperatures. This may be particularly important, as a large local temperature variability helps ensuring that all but the coldest microhabitats will persist under climate warming (Scherrer and Körner 2011, Suggitt et al. 2018). Temperature changes after decades of climate warming are therefore likely to be small as compared to temperature variations among microsites with different topography and hydrology, making it possible for species to escape the effects of climate change by relocating locally (Opedal et al. 2015, Jiménez-Alfaro et al. 2024).

Our results point to a small direct effect of climatic variability on plant community composition and species richness along the Nuuk fjord, but increasing temperatures may still have indirect long-term effects on the vegetation if they result in increasing shrub densities and competitive exclusion of small species. The higher growth rates observed for shrubs in the Nuuk area over the past decades (Prendin et al. 2022) suggests that shrubs will become increasingly dominant in coming years. As there is a slight tendency for species richness to decrease at high cover, our results support the notion that competition from warm-adapted species may affect Arctic plant diversity (Elmendorf et al. 2012b, Robinson et al. 2018, Oke et al. 2023).

Predicting the long-term impact of climate change is, however, complicated by its interaction with a range of other factors, including herbivory. Results from warming experiments have demonstrated that presence of large herbivores can help preventing competitive exclusion of rare species and thereby contribute to the maintenance of diversity under future climatic conditions (Post 2013, Kaarlejärvi et al. 2017, Post et al. 2022). Trampling and grazing of herbivores can also reduce the depth of the moss layer and thereby affect soil temperatures and the composition of plant communities (Van Der Wal et al. 2001, Brooker and Van Der Wal 2003). Particularly seed establishment may depend on microsites with favourable temperatures (Graae et al. 2011), and in the absence of herbivory such microsites might disappear more rapidly as the climate becomes warmer.

The weak influence of temperature in our study suggests that broad-scale climatic differences have played a minor role for plant communities in the past, but future climate change is likely to have both direct and indirect effects that are fundamentally different. Most importantly, the expected 3.3–10°C warming over the remainder of the century (AMAP 2021) will presumably cause warm-adapted species to become increasingly dominant, even in many areas where species adapted to a short snow-free season have had a competitive advantage till now. In our study several of the species that are often associated with late-melting snow beds, including *Huperzia selago, Harrimanella hypnoides*, *Cassiope tetragona* and *Oxyria digyna* (Böcher 1954, Kelly and Funder 1974, Callaghan et al. 1989) were exclusively found in plots with < 80% cover and predominately at high elevations. This underlines the pivotal role that late melting snow beds may have for preventing competitive exclusion of many small species in the Nuuk area and suggests that local variability in plant community composition is partly related to factors that locally prevent large shrubs from outcompeting small species. An early disappearance of snow cover may therefore have a large impact on plant communities. This influence can be further exacerbated by the change in snow cover that will result from an increasing prevalence of extreme weather events in future (Harris et al. 2018, Frei and Henry 2022). Although herbivory can prevent warm-adapted species from outcompeting those that tolerate harsher conditions (Post 2013, Kaarlejärvi et al. 2017), one likely outcome of increasing temperatures is therefore that the cold-adapted species are forced to survive in new areas as these become climatically suitable. Jointly these indirect effects of climate change are likely to have a considerable impact on the low Arctic tundra.

Our study covers one of the largest coast-to-inland climatic gradients in Greenland, yet the differences in temperatures and plant community composition along the fjord are minor (Fig. 4, Supporting information). Instead, a large proportion of the variation in plant communities occurs over small spatial scales. If the fine-scale environmental variability that helps maintain this community variability diminishes with climate change, increasing dominance of warm-adapted shrubs may lead to competitive exclusion of some of the small-statured species, at least locally. As the fjord system covers climatic gradients that resemble those observed in many other parts of western Greenland, our study helps painting a broad picture of how the low Arctic tundra vegetation will develop in response to changing climatic conditions.

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Author contributions

Jacob Nabe-Nielsen: Conceptualization (lead); Formal analysis (equal); Investigation (equal); Methodology (lead);

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Visualization (lead); Writing – original draft (lead); Writing – review and editing (equal). **Louise I. Nabe-Nielsen**: Investigation (equal); Writing – review and editing (equal). **Otso Ovaskainen**: Formal analysis (equal); Visualization (supporting); Writing – review and editing (equal).

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Data availability statement

Data are available from the Dryad Digital Repository: https://doi.org/10.5061/dryad.8sf7m0d17 (Nabe-Nielsen et al. 2025).

Supporting information

The Supporting information associated with this article is available with the online version.

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